Google Cloud

Overview of Google's cloud Data analytics tool

As the cloud continues becoming a more prominent reach in the data world, all kinds of exciting new data tools will arise.

As a data professional, it's important to know that cloud data tools will usually be specific to each stage of the data lifecycle.

And there will probably be a lot of choices for each stage.

So, how do you choose the right solution for a particular task?

In this video, we'll introduce you to some of the tools offered by Google and explore how they can support many types of projects and initiatives.

First is BigQuery, a serverless data warehouse.

This tool is pretty popular in the data world because it works across many major cloud platforms.

With BigQuery, you use SQL to query and organize data.

There are also some really cool machine learning and artificial intelligence tools integrated within the platform.

And BigQuery includes built-in business intelligence engines to create interactive and responsive data insights.

As a data analyst, you may use BigQuery to process seemingly unlimited amounts of data — think terabytes and even more terabytes— without having to manage the database.

Another solution that may become a part of your toolkit is Looker.

Looker organizes business data and builds workflows and applications based on data insights.

But it's primarily a data visualization and reporting tool.

As a data analyst, you might use Looker to integrate various file types like CSV, JSON, and Excel into a single application.

You may also use it to publish data in a variety of dashboards.

Next is Dataproc.

This service is fully managed and allows you to run Apache Hadoop, Apache Spark, and Apache Flink, along with many other open-source tools and frameworks.

Dataproc can modernize data lakes and perform ETL functions —or extract, transform, and load— for massive amounts of data.

As a data professional, you might use Dataproc to process large data workloads and place them in clusters that allow you to access your data quickly.

Then, there's Dataflow, which gives you the ability to stream and batch process data in a serverless application.

Data professionals may use Dataflow to develop data processing pipelines.

It can assist with reading the data from its original source, transforming it, and writing it back to your database.

You may also work with Cloud Data Fusion, another fully managed service that allows you to integrate multiple datasets of any size.

As a cloud data analyst, you might use Cloud Data Fusion because it allows you to use a graphical user interface instead of code to manage your data pipelines.

Dataplex also allows you to work with multiple data sources.

It creates a central hub for managing and monitoring data to work across various data lakes, data warehouses, and data marts.

As a data analyst, you may access Dataplex with BigQuery.

This will allow you to access data from a variety of databases and platforms with a single interface.

Finally, BigLake is a storage engine that you can use to unify data warehouses and lakes with BigQuery and open-source frameworks.

It also provides options for access control, and multi-cloud storage.

These are just some of the standard tools you might encounter as a data analyst, and there are even more on the horizon.